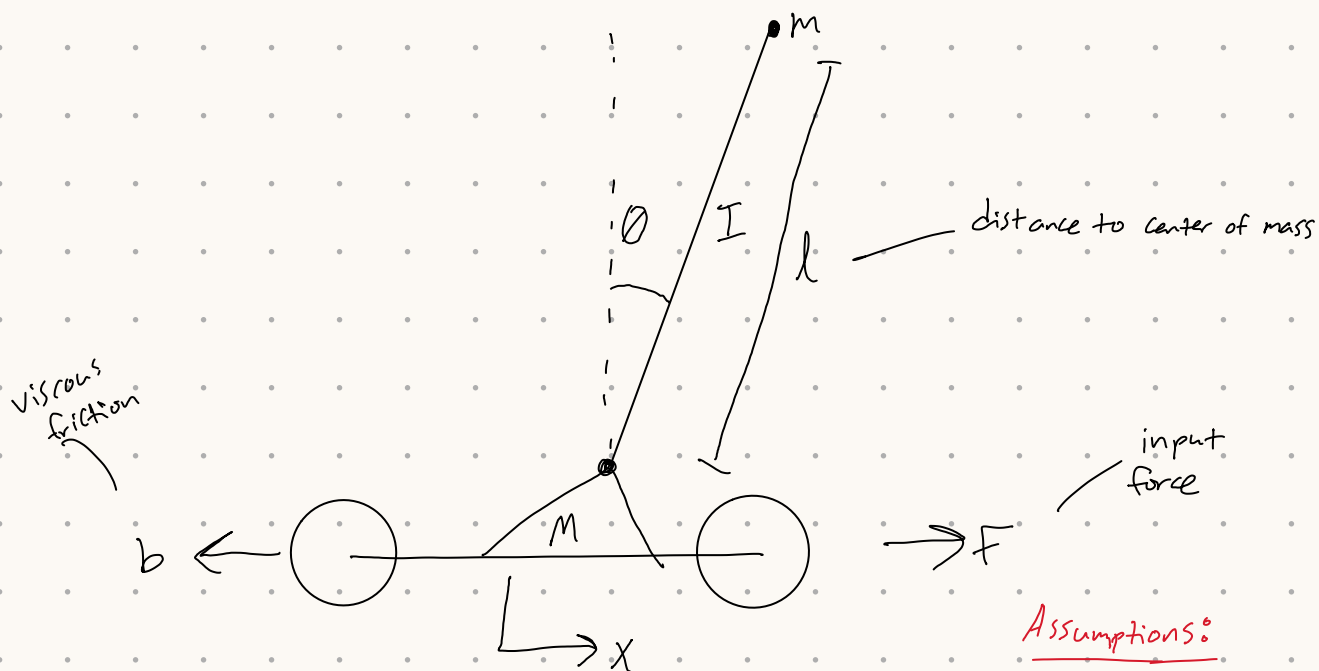




Assumptions:

- Small angle theorem
- all frictional forces can be modeled with viscous friction

$$(M+m)\ddot{x} + b\dot{x} + ml\ddot{\theta} = F \quad (1)$$

$$(I+ml^2)\ddot{\theta} + mgl\theta + ml\ddot{x} = 0 \quad (2)$$

State-space form:

Let $q = \underbrace{(M+m)(I+ml^2)}_B - \underbrace{(ml)^2}_B$

$$\begin{bmatrix} \dot{x} \\ \ddot{x} \\ \dot{\theta} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & \frac{-(I+ml^2)b}{q} & \frac{-(ml)^2g}{q} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \frac{-mlb}{q} & \frac{mgl(M+m)}{q} & 0 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{I+ml^2}{q} \\ 0 \\ \frac{ml}{q} \end{bmatrix} F$$

Motor current as a control input:

$$T_{\text{motor}} = K_t i$$

$$T_{\text{out}} = G T_m$$

$$F = \frac{G K_t}{r} i = \frac{T_{\text{out}}}{r} i$$

↑
input force

multiply B by

$$\frac{G K_t}{r}$$

$$\begin{bmatrix} 0 \\ \frac{G K_t (I+ml^2)}{q r} \\ 0 \\ \frac{G K_t m l}{q r} \end{bmatrix} i$$

Motor Voltage as a Control Input:

$$i = \frac{V - V_{\text{back EMF}}}{R}$$

motor winding resistance

$$V_{\text{EMF}} = K_e \omega_m, \quad \omega_m = \frac{G \dot{x}}{r}$$

$$\Rightarrow F = \frac{G K_t}{r} \left(\frac{V - K_e \frac{G \dot{x}}{r}}{R} \right)$$

$$\Rightarrow F = \underbrace{\left(\frac{G K_t}{r R} \right)}_{\text{input gain}} V - \underbrace{\left(\frac{G^2 K_t K_e}{r^2 R} \right)}_{\text{damping effect}} \dot{x}$$

multiply to matrix B

replace b in matrix A:

$$b_{\text{eff}} = b + \frac{G^2 K_t K_e}{r^2 R}$$

Parameters:

$$M = 2.525 \text{ kg}$$

$$m = 0.152 \text{ kg}$$

$$l = 0.105 \text{ m}$$

$$I = 3.770 \text{ e-}3$$

$$r = 0.038 \text{ m}$$

$$G = 3$$

$$R = 6.5 \Omega$$

$$K_t = 0.1394$$

$$K_e = 0.1394$$

$$b =$$

$$g = 9.81 \text{ m/s}^2$$

needs updating if we add weight to pendulum

Determining R of brushed motor windings:

Using a current source of 1A, stall the motor and measure motor voltage. Average this value over multiple shaft angles.

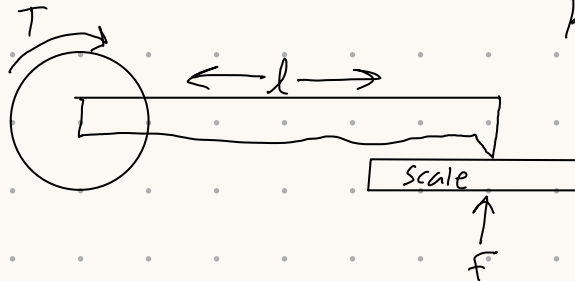
$$V = IR \quad \Rightarrow \quad R = \frac{V}{I} = \frac{V}{1A} \Rightarrow R = V$$

Determining K_t and K_e

In SI units, $K_t = K_e$

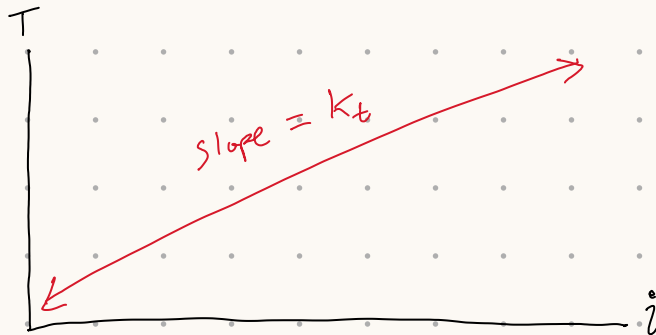
$$T = Fl$$

Lever arm method:



$$K_t = \frac{T}{i} = \frac{Fl}{i}$$

measure at multiple shaft angles and motor currents



Sanity check: $\omega [\text{rad/s}] = \frac{V_{\text{applied}}}{K_t}$

$$\omega = \text{RPM} \cdot \frac{2\pi}{60}$$

if error $< 5\%$, LQR control should be accurate enough neglecting b

Finding damping coefficient b :

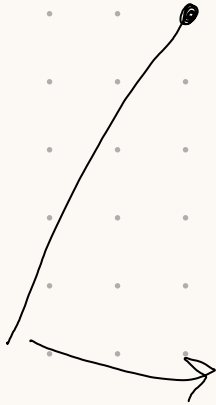
$$b = \frac{K_t (V_{\text{applied}} - K_t \omega_{\text{actual}})}{R \omega_{\text{actual}}}$$

check at multiple V_{applied} to ensure you get a good or at least average value. (e.g., if b is non-linear)

Finding rotational inertia of pendulum:

$$I_{\text{pendulum}} = \frac{mglT^2}{4\pi^2}$$

T is period [seconds]



Let the pendulum swing freely
and measure the period

$$T = 0.975 \text{ seconds}$$

Finding Drivetrain rotational inertia: !! Use SI units !!

replace M with M_{eff}

$$M_{\text{eff}} = M + \frac{JG^2}{r^2}$$

- apply a step current input to the motor, with the wheels free to spin in the air.
- Monitor the voltage over time.
- Find the slope of the linear portion (viscous friction ≈ 0)

$$\alpha = \frac{\text{Voltage slope}}{K_e}$$

$$J = \frac{K_t \dot{i}_{\text{applied}}}{\alpha}$$

in SI units, $K_t = K_e$

$$\Rightarrow J = \frac{\text{Voltage slope}}{\dot{i}_{\text{applied}}}$$

Calculating Matrices for Voltage Control

$$\begin{bmatrix} \dot{X} \\ \ddot{X} \\ \dot{\theta} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\frac{I+ml^2}{q} \cdot \frac{G^2 k_t k_e}{r^2 R} & -\frac{(ml)^2 g}{q} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & +\frac{ml}{q} \cdot \frac{G^2 k_t k_e}{r^2 R} & \frac{mgl(M+m)}{q} & 0 \end{bmatrix} X + \begin{bmatrix} 0 \\ \frac{I+ml^2}{q} \cdot \frac{G k_t}{r R} \\ 0 \\ -\frac{ml}{q} \cdot \frac{G k_t}{r R} \end{bmatrix} V$$

$$q = (M+m)(I+ml^2) - (ml)^2 = 14.32 e^{-3}$$

$$\begin{bmatrix} \dot{X} \\ \ddot{X} \\ \dot{\theta} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -7.084 & -174.5 e^{-3} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 20.76 & 29.26 & 0 \end{bmatrix} \begin{bmatrix} X \\ \dot{X} \\ \theta \\ \dot{\theta} \end{bmatrix} + \begin{bmatrix} 0 \\ 643.7 e^{-3} \\ 0 \\ -1.887 \end{bmatrix} V$$